THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** the mass of the ball and its speed and direction	these decide whether the ball crosses the boundary
2	**(d)** Using the speed and angle of a kick to find how far a football will travel	reasoned expectation from evidence
3	**(b)** sublimation	camphor sublimes, sand does not
4	**(b)** distillate	collected in a separate vessel
5	**(c)** 8.8 g	$\frac{44}{12} \times 2.4 = 8.8$
6	**(b)** $CaCO_3$	Ca^{2+} and CO_3^{2-}
7	**(c)** 14	neutrons = 27 − 13
8	**(c)** $v = s/t^2$	not valid in general
9	**(d)** constant velocity, zero acceleration	slope is zero
10	**(c)** 4 N	add if same direction; subtract if opposite
11	**(b)** increase friction	grip
12	**(d)** 50000 J	$\frac{1}{2}mv^2$
13	**(b)** 1700 m	d = v × t
14	**(a)** 0.2 mm	1 mm = 1000 µm
15	**(b)** a well-defined nucleus	nucleoid region
16	**(b)** phloem	translocation
17	**(a)** chitin	absorptive nutrition
18	**(a)** fruit	ovule becomes the seed
19	**(b)** bread mould (Rhizopus)	light spores in air
20	**(a)** 3×10^8 m s^{-1}	c

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	e.g. What was the humidity the last time it rained? Is the temperature dropping now? What are today's wind speed and direction? (1+1).	1+1
22	Molecular mass is for covalent compounds made of discrete molecules (1); formula unit mass is for ionic compounds, using the simplest formula (1).	1+1
23	Uniform: a train at constant speed on a straight track; non-uniform: a car speeding up from a traffic signal (1+1).	1+1
24	Net force is zero, so by the first law it continues at constant velocity if already moving (2).	2
25	Chloroplasts (photosynthesis), chromoplasts (colour), leucoplasts (storage) (2).	2
26	Protects the heart and lungs and expands and contracts during breathing (2).	2
27	So that the zygote restores the full number after fertilisation (2).	2
28	A layer of ozone in the stratosphere (1); it absorbs harmful UV radiation (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Gas particles move freely in all directions and mix uniformly (1.5); example: air or H_2 + O_2; heterogeneous: smoke/fog (1.5).	1½+1½
30	Atomic number 11 for both; mass numbers 23 and 24 (2); same element — they are isotopes (1).	2+1
31	Real machines lose energy to friction and air resistance as heat and sound, so they slow down; energy cannot be created (3).	3
32	Unwanted or harmful sound (1); sleep disturbance, hearing loss (1); sound-proofing, limiting volume (1).	1+1+1

33	Receives stimuli and transmits impulses (2); neuron (1).	2+1
34	They have needle-like leaves that reduce water loss (1.5); they do not need water for fertilisation and form seeds (1.5).	1½+1½
35	Al^{3+}, O^{2-} → Al_2O_3 (1); $CaCO_3$ (1); Na_2SO_4 (1).	1+1+1
	OR: Strong electrostatic attraction between ions (1); regular lattice needs much energy to melt (1); displacement of layers brings like charges together causing brittleness (1).	1+1+1
36	Uneven heating of the equator and poles produces low and high pressure belts (1.5); air flows from high to low pressure belts (1.5).	1½+1½
	OR: Winds, differences in temperature and salinity, Earth's rotation (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Energy released when nuclei split or combine; used in electricity generation, medicine, research (2). (ii) Raja Ramanna (1). (iii) Heat from nuclear reactions makes steam that drives turbines (2).	2+1+2
	OR: (i) CO_2 and NH_3 covalent; NaCl and $MgCl_2$ ionic (1). (ii) 44 u and 17 u (1). (iii) 58.5 u and 95 u (1). (iv) Ionic solids are lattices of ions without discrete molecules, so the simplest formula is used (2).	1+1+1+2
38	(i) First: lurch in a bus; second: F = ma for a car; third: rocket launch (3). (ii) First law (inertia) – the body continues moving forward (2).	3+2
	OR: (i) Treat both and the string as one object; internal tension is ignored, $a = F/(m_1 + m_2)$ (2). (ii) 2 m s^{-2} (1.5). (iii) T = m_2 a = 6 N when box 1 is pulled (1.5).	2+1½+1½
39	(i) Chloroplasts, chromoplasts, leucoplasts with functions (2). (ii) Own DNA and ribosomes (1). (iii) No; roots have leucoplasts for storage (2).	2+1+2
	OR: (i) They contain actively dividing cells (1). (ii) Cell division (mitosis) (1). (iii) Mitosis and meiosis (1). (iv) Growth, repair, replacement and asexual reproduction (2).	1+1+1+2
40	(i) It forms gametes with half the chromosome number (1). (ii) 23, 23 and 46 (2). (iii) Each gamete receives a random combination of chromosomes from each pair (2).	1+2+2
	OR: (i) Embryo, foetus growth, readiness for birth (3). (ii) Nourishment, oxygen and waste removal (1). (iii) Smoking and alcohol (1).	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER